If we had allowed vertex v_4 to move both vertically and horizontally, we would have had, instead, a 3D chart showing a surface.

This is shown in figure 16.8 for complete graph K_5, or to be more precise, for one of the possible embeddings for this graph.

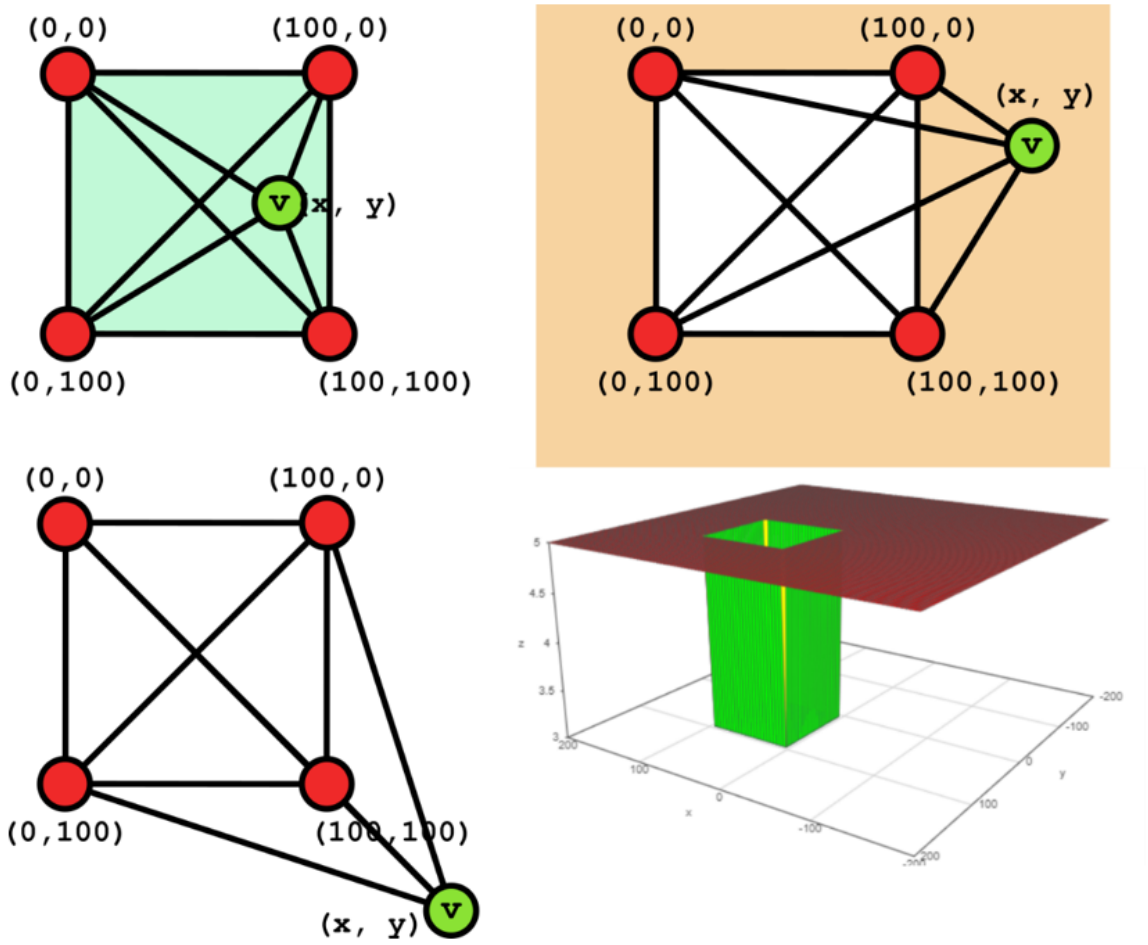


Figure 16.8 The cost function for "number of intersections" for a specific embedding of K_{45}. The position of 4 vertices is locked, and only vertex v can be moved freely. If we move v inside the perimeter of square created by the other vertices, the number of intersections is consistently 3; outside of the square, it becomes 5. Notice that here we have surfaces of discontinuity!